Kelvin Siu

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EDUCATION

University of Nevada, Las Vegas GPA 3.2
Bachelor of Science in Computer Science July 2026

WORK EXPERIENCE

Software Development Intern, VortexNet June 2026 – September 2026

- Built an end-to-end **AI interview platform** supporting **real-time voice interviews**, coding assessments, system-design whiteboarding, and **asynchronous workflows** through integrated LLM agents.
- Designed a low-latency fault-tolerant voice pipeline using bidirectional **WebSockets**, streaming **ASR/TTS**, fallback mechanisms, supporting ~1,000 concurrent connections with ~2s P99 voice latency.
- Developed coding and system-design evaluation workflows with sandboxed execution and **multimodal LLM analysis**, implementing execution timeouts, bounded polling, content-hash caching, and persistent storage to improve evaluation reliability and reproducibility.
- Improved scalability by decoupling report generation into asynchronous **SQS** workers with retry mechanisms and dead letter queues and deploying services on **EKS with autoscaling** and **CloudWatch monitoring**.

Full Stack Web Development Intern, DataX UNLV January 2026 – May 2026

- Developed a bioinformatics platform that runs deep-learning enzyme prediction models, enabling researchers to submit protein sequences and inspect EC-number predictions, confidence scores, and biological evidence.
- Engineered a secure password recovery service for 100+ registered researchers using **MySQL** for account and token storage; integrated **Google SMTP** to send password-reset emails with time-limited reset links, validating tokens server-side before updating credentials with **bcrypt** hashing.
- Built visitor analytics that captures request metadata and integrates a free IP geolocation **API** to map visitor IP addresses to approximate countries, aggregating the results to help understanding geographic platform reach.

PROJECTS

AI D&D Platform Spring 2026

- Built an end-to-end multiplayer AI D&D platform with graphical character creation, live sessions, an AI game host that generates dynamic stories and supports interactive gameplay.
- Engineered a server-authoritative AI pipeline using **ASP.NET Core, Supabase PostgreSQL, OpenAI,** and **pgvector**, with **JWT/RLS security** and validated LLM outputs to protect multiplayer game state.
- Reduced LLM context usage by ~90% through pgvector-based retrieval of relevant campaign memories instead of full conversation history, while deploying the containerized platform on Azure Container Apps.

AI Meeting Action Agent Spring 2026

- Built an AI meeting workspace that turns audio, documents, and transcripts into summaries, decisions, risks, action items, evidence-backed Q&A, and downloadable reports.
- Designed a **RAG**-based pipeline using **LangChain**, **OpenAI embeddings**, and **pgvector** to semantically retrieve relevant transcript chunks for grounding LLM responses in evidence to improve answer accuracy.
- Implemented **Kafka**-backed worker architecture to asynchronously process meeting-analysis jobs, with worker health monitoring to improve throughput and reliability.

LANGUAGES & TECHNOLOGY

Languages: C++ | Python | TypeScript | JavaScript | SQL | HTML | CSS
Frameworks & Libraries: FastAPI | ASP.NET Core | React | Next.js | LangChain | LangGraph
AI & Backend: OpenAI API | RAG | pgvector | REST APIs | WebSockets | Judge0
Databases: PostgreSQL | MySQL | Supabase | DynamoDB
Cloud & DevOps: AWS | AWS S3 | SQS | Azure | Docker | Kubernetes | EKS | CI/CD | Git | GitHub
Messaging & Security: Kafka | JWT | Row-Level Security | SMTP
Certification: AWS Certified Cloud Practitioner, Amazon Web Services